

Gradient of a scalar

$\phi \rightarrow \text{scalar.}$

$$\therefore \nabla = \text{del operator} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

Gradient is represented by $\therefore d = \text{differential operator} = \frac{\partial}{\partial x} dx + \frac{\partial}{\partial y} dy + \frac{\partial}{\partial z} dz$

" $\nabla \phi$ "

$$\nabla \phi = \left(\frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right)$$

$$\boxed{\nabla \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}} \quad \text{"gradient" of scalar}$$

Example: If $\phi = (x, y, z) = 3x^2y - y^3z^3$ find $\nabla \phi$ at point $(1, -2, -1)$.

$$\nabla \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}$$

$$\nabla \phi = \frac{\partial}{\partial x} (3x^2y - y^3z^3) \hat{i} + \frac{\partial}{\partial y} (3x^2y - y^3z^3) \hat{j} + \frac{\partial}{\partial z} (3x^2y - y^3z^3) \hat{k}$$

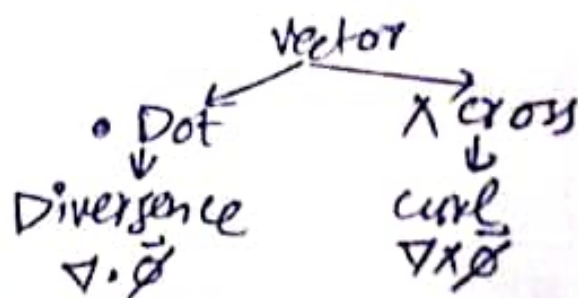
$$6xy \hat{i} + (3x^2 - 3y^3z^3) \hat{j} - 2y^3z^3 \hat{k}$$

$$(\nabla \phi)_{(1, -2, -1)} = -12 \hat{i} - 9 \hat{j} - 16 \hat{k}$$

Divergence of vector:-

$\vec{\phi} \rightarrow \text{vector.}$

$$\vec{\phi} = \phi_1 \hat{i} + \phi_2 \hat{j} + \phi_3 \hat{k}$$



$$\nabla \cdot \vec{\phi} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (\phi_1 \hat{i} + \phi_2 \hat{j} + \phi_3 \hat{k})$$

$$\boxed{\nabla \cdot \vec{\phi} = \frac{\partial \phi_1}{\partial x} + \frac{\partial \phi_2}{\partial y} + \frac{\partial \phi_3}{\partial z}} \rightarrow \text{div of vector!}$$

Example: If $\vec{A} = x^2 \hat{i} - 2y^3 z^2 \hat{j} + xy^2 z \hat{k}$ find $\nabla \cdot \vec{A}$ at point $(1, -1, 1)$.

$$\nabla \cdot \vec{\phi} = \frac{\partial \phi_1}{\partial x} + \frac{\partial \phi_2}{\partial y} + \frac{\partial \phi_3}{\partial z}$$

$$\phi_1 = x^2 z, \quad \phi_2 = -2y^3 z^2$$

$$\phi_3 = xy^2 z$$

$$\nabla \cdot \vec{\phi} = \frac{\partial (x^2 z)}{\partial x} + \frac{\partial (-2y^3 z^2)}{\partial y} + \frac{\partial (xy^2 z)}{\partial z}$$

$$\nabla \cdot \vec{\phi} = 2xz - 6y^2 z^2 + xy^2$$

$$\nabla \cdot \vec{\phi}_{(1, -1, 1)} = -3$$

curl of vector

$$\vec{\phi} = \phi_1 \hat{i} + \phi_2 \hat{j} + \phi_3 \hat{k}$$

$$\nabla \times \vec{\phi} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \phi_1 & \phi_2 & \phi_3 \end{vmatrix}$$

$$\nabla \times \vec{\phi} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \times (\phi_1 \hat{i} + \phi_2 \hat{j} + \phi_3 \hat{k})$$

$$\boxed{\nabla \times \vec{\phi} = \hat{i} \left(\frac{\partial \phi_3}{\partial y} - \frac{\partial \phi_2}{\partial z} \right) - \hat{j} \left(\frac{\partial \phi_3}{\partial x} - \frac{\partial \phi_1}{\partial z} \right) + \hat{k} \left(\frac{\partial \phi_2}{\partial x} - \frac{\partial \phi_1}{\partial y} \right)}$$

curl of vector

Example: If $\vec{\phi} = xz^3\hat{i} - 2x^2yz\hat{j} + 2yz^4\hat{k}$ find $\nabla \times \vec{\phi}$ at $(1, -1, 1)$

$$\nabla \times \vec{\phi} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz^3 & -2x^2yz & 2yz^4 \end{vmatrix}$$

$$\begin{aligned} \phi_1 &= xz^3 \\ \phi_2 &= -2x^2yz \\ \phi_3 &= 2yz^4 \end{aligned}$$

$$\begin{aligned} \nabla \times \vec{\phi} &= \hat{i} \left(\frac{\partial (2yz^4)}{\partial y} - \frac{\partial (-2x^2yz)}{\partial z} \right) - \hat{j} \left(\frac{\partial (2yz^4)}{\partial x} - \frac{\partial (xz^3)}{\partial z} \right) \\ &\quad + \hat{k} \left(\frac{\partial (-2x^2yz)}{\partial x} - \frac{\partial (xz^3)}{\partial y} \right) \\ &= (2z^4 + 2x^2y)\hat{i} + 3xz^2\hat{j} - 4xyz\hat{k} \end{aligned}$$

$$\nabla \times \vec{\phi} = 3\hat{j} + 4\hat{k} \text{ at } (1, -1, 1)$$

Solenoidal

If $\nabla \cdot \vec{\phi} \text{ (Div)} = 0$
 $\vec{\phi} \rightarrow \text{solenoidal}$

Irrrotational

If $\nabla \times \vec{\phi} \text{ (curl)} = 0$
 $\vec{\phi} \rightarrow \text{Irrrotational}$

Questions:

(Related questions to gradient, Divergence & curl)
Q# If $\phi = x^n + y^n + z^n$ show that $\vec{r} \cdot \nabla \phi = n\phi$.

As we know

$$\nabla \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}$$

$$\phi = x^n + y^n + z^n$$
$$= \frac{\partial (x^n + y^n + z^n)}{\partial x} \hat{i} + \frac{\partial (x^n + y^n + z^n)}{\partial y} \hat{j} + \frac{\partial (x^n + y^n + z^n)}{\partial z} \hat{k}$$
$$= nx^{n-1} \hat{i} + ny^{n-1} \hat{j} + nz^{n-1} \hat{k}$$

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\vec{r} \cdot \nabla \phi = (x\hat{i} + y\hat{j} + z\hat{k}) \cdot (nx^{n-1}\hat{i} + ny^{n-1}\hat{j} + nz^{n-1}\hat{k})$$
$$= nx^n + ny^n + nz^n = n(x^n + y^n + z^n)$$
$$\therefore \phi = x^n + y^n + z^n$$

$$\vec{r} \cdot \nabla \phi = n\phi \quad \text{proved.}$$

Q# If $\phi = (x^2 + y^2 + z^2) e^{-\sqrt{x^2 + y^2 + z^2}}$ find $\nabla \phi$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$r^2 = x^2 + y^2 + z^2$$

put values of r ϕ becomes
 $\phi = r^2 e^{-r}$ Now find $\nabla \phi$.

$$\nabla \phi = \frac{\partial \phi}{\partial r}$$

$$= \frac{\partial (r^2 e^{-r})}{\partial r} \Rightarrow \text{product rule.}$$

$$\nabla \phi = -r^2 e^{-r} + e^{-r} 2r \Rightarrow e^{-r} (-r^2 + 2r)$$

Q# Determine the constant a so that $\vec{v} = (x+3y)\hat{i} + (y-2x)\hat{j} + (x+az)\hat{k}$ is solenoidal.

solenoidal, $\nabla \cdot \vec{v} = 0$ — (A)

$$\nabla \cdot \vec{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

$$v_1 = x+3y, v_2 = y-2x$$

$$v_3 = x+az$$

$$\nabla \cdot \vec{v} = \frac{\partial}{\partial x}(x+3y) + \frac{\partial}{\partial y}(y-2x) + \frac{\partial}{\partial z}(x+az)$$

$$\nabla \cdot \vec{v} = 1 + 1 + a$$

$$\nabla \cdot \vec{v} = 2 + a \quad \text{put in (A)}$$

$$\nabla \cdot \vec{y} = 0$$

Q#

$$2 + a = 0$$

$$\Rightarrow \boxed{a = -2}$$

Q# of $A = x^2y\hat{i} - 2xz\hat{j} + 2yz\hat{k}$ find $\nabla \times \nabla \times A$

$\nabla \times \vec{A} = \text{curl of } A =$

$$= \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \hat{i} - \left(\frac{\partial A_3}{\partial x} - \frac{\partial A_1}{\partial z} \right) \hat{j} + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \hat{k}$$

$$A_1 = x^2y, \quad A_2 = -2xz, \quad A_3 = 2yz$$

$$= \left(\frac{\partial (2yz)}{\partial y} - \frac{\partial (-2xz)}{\partial z} \right) \hat{i} - \left(\frac{\partial (2yz)}{\partial x} - \frac{\partial (x^2y)}{\partial z} \right) \hat{j} +$$

$$\left(\frac{\partial (-2xz)}{\partial x} - \frac{\partial (x^2y)}{\partial y} \right) \hat{k}$$

$$\nabla \times \vec{A} = (2x + 2z) \hat{i} - (x^2 + 2z) \hat{k}$$

Now again curl.

$$\nabla \times \nabla \times A =$$

$$\nabla \times A = (2x + 2z) \hat{i} - (x^2 + 2z) \hat{k}$$

$$A_1 = 2x + 2z, \quad A_3 = -(x^2 + 2z)$$

Again $A_2 = 0$
Put values in previous above formulae

$$\nabla \times \nabla \times A = (2x + 2) \hat{j}$$

Theorem:-

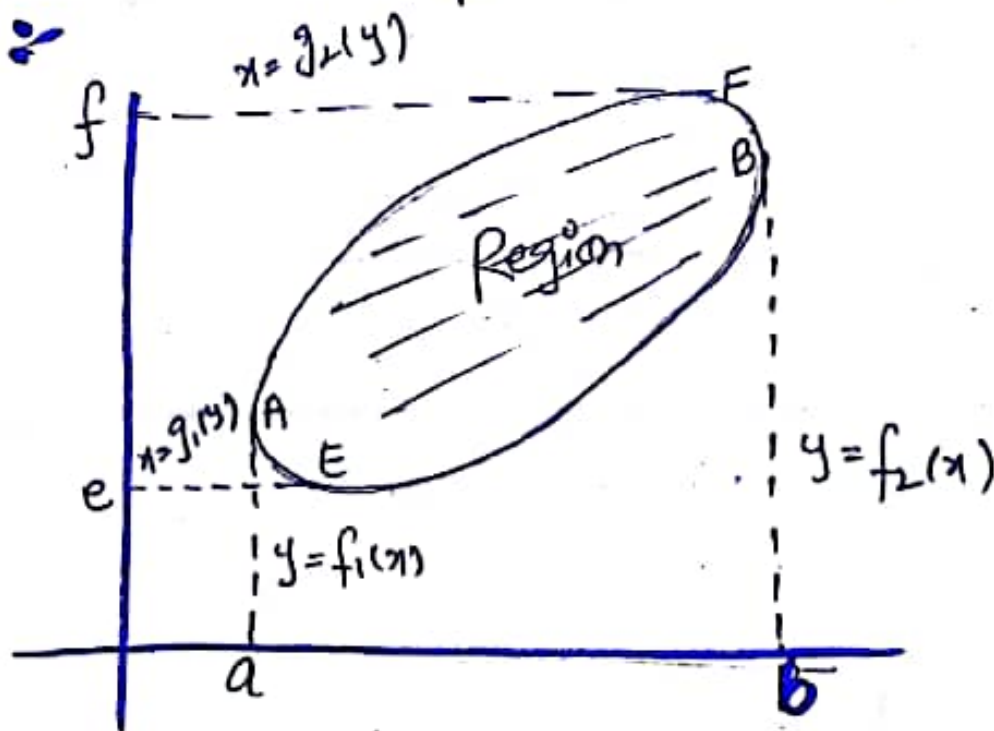
Green theorem:

Q# state and prove green theorem:

let R be a simply connected closed region and if $M(x,y)$ and $N(x,y)$ are continuous functions in xy plane having partial derivatives in R , then

$$\oint_C (M dx + N dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Proof:-



let us we have two curve AEB and ~~BAF~~. By R.H.S "For curve AEB"

$$\iint_R \frac{\partial M}{\partial y} dx dy = \int_{f_1(x)}^{f_2(x)} \int_a^b \frac{\partial M}{\partial y} dx dy$$

* (variable will integrate 1st we will integrating Limits)*

$$\int_a^b \int_{f_1(x)}^{f_2(x)} \frac{\partial M}{\partial y} dy dx$$

* (partial will cancel out by integrated as derivation and integration are reverse process of each other)*

$$\iint_R \frac{\partial M}{\partial y} dx dy = \int_a^b M(x, y) \Big|_{f_1(x)}^{f_2(x)} dx$$

Put limits in 'y'

$$= \int_a^b M(x, f_2(x)) dx - \int_a^b M(x, f_1(x)) dx$$

$$= - \int_a^b M(x, f_1(x)) dx + \int_a^b M(x, f_2(x)) dx$$

$$\therefore \int_a^b f_1(x) dx = - \int_b^a f_1(x) dx$$

$$- \int_a^b M(x, f_1(x)) dx = - \int_b^a M(x, f_2(x)) dx$$

$$-\left(\int_a^b M(x, f_2(x)) dx + \int_a^b M(x, f_1(x)) dx\right)$$

Formula of closed interval.

$$= - \oint_C M dx$$

Now

$$\iint_R \frac{\partial M}{\partial y} dx dy = - \oint_C M dx$$

$$\oint_C M dx = - \iint_R \frac{\partial M}{\partial y} dx dy \quad \text{--- (A)}$$

Now for curve EAF:

$$\iint_R \frac{\partial N}{\partial x} dx dy = \int_e^f \int_{g_1(y)}^{g_2(y)} \frac{\partial N}{\partial x} dx dy$$

(partial will be cancel out by integral).

$$= \int_e^f N(x, y) \Big|_{g_1(y)}^{g_2(y)} dy$$

$$\int_e^f N(g_2(y), y) dy - \int_e^f N(g_1(y), y) dy$$

$$\therefore \int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$= \int_e^f N(g_2(y), y) dy + \int_f^e N(g_1(y), y) dy$$

$$= \int_f^e N(g_1(y), y) dy + \int_e^f N(g_2(y), y) dy$$

definition of closed interval.

$$\boxed{\iint_R \frac{\partial N}{\partial x} dx dy = \oint_C N dy}$$

$$\oint_C N dy \stackrel{\text{or}}{=} \iint_R \frac{\partial N}{\partial x} dx dy \rightarrow \textcircled{B}$$

By adding $\textcircled{A}$ and $\textcircled{B}$

$$\oint_C M dx + \oint_C N dy = - \iint_R \frac{\partial M}{\partial y} dx dy + \iint_R \frac{\partial N}{\partial x} dx dy$$

$$\oint_C (Mdx + Ndy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

proved -

Q# verify Green's theorem in plane for $M = xy + y^2$ and $N = x^2$ where C is the closed curve of the region bounded by $y = x$ and $y = x^2$.

$$\iint_R \underbrace{\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy}_{\text{L.H.S.}} = \oint_C \underbrace{M dx + N dy}_{\text{R.H.S.}} \quad \text{--- (1)}$$

$$\text{R.H.S.} = \oint_C M dx + N dy = \oint_C (xy + y^2) dx + x^2 dy$$

There are two curves $C_1: y = x$ and $C_2: y = x^2$

$$\text{Now } y = x$$

$$y = x^2$$

$$x = x^2$$

$$x - x^2 = 0$$

$$x(x-1) = 0$$

$$\boxed{x = 0, 1}$$

For curve C_1 $y = x \Rightarrow dy = dx$

put value in (1).

$$\oint_C (xy + y^2) dx + x^2 dy$$

$$\int_0^1 ((x)(x) + x^2) dx + x^2 dx$$

$$\int_0^1 3x^2 dx \Rightarrow \frac{3x^3}{3} \Big|_0^1 = 1$$

$$\frac{2x^3}{3} \Big|_1^0 \Rightarrow 0 - 1 \Rightarrow \boxed{-1}$$

Now $C_2: y = x^2$

$\frac{dy}{dx} = 2x \Rightarrow dy = 2x dx$

Put in (1)

$$\int_C (xy + y^2) dx + x^2 dy$$

$$\int_0^1 (x \cdot x^2 + (x^2)^2) dx + x^2 (2x dx)$$

$$\int_0^1 (x^3 + x^4) dx + 2x^3 dx$$

$$\int_0^1 (3x^3 + x^4) dx \Rightarrow \frac{3}{4} x^4 + \frac{x^5}{5} \Big|_0^1$$

$$\frac{3}{4} + \frac{1}{5} = \boxed{\frac{19}{20}}$$

From both curves.

$$C_1 + C_2 = -1 + \frac{19}{20} \Rightarrow -\frac{1}{20}$$

$$\boxed{\int_C (M dx + N dy) = -\frac{1}{20} \rightarrow \textcircled{A}}$$

Now L.H.S / B

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy \rightarrow (2)$$

$$\frac{\partial N}{\partial x} = \frac{\partial x^2}{\partial x} = 2x, \quad \frac{\partial M}{\partial y} = \frac{\partial (xy + y^2)}{\partial y} = x + 2y$$

Put in (2)

$$\int_{x^2}^x \int_0^1 (2x - (x + 2y)) dx dy$$

$$\int_{x^2}^x \int_0^1 (x - 2y) dx dy$$

$$\int_0^1 \int_{x^2}^x (x - 2y) dy dx$$

$$\int_0^1 \left[xy - \frac{2y^2}{2} \right]_{x^2}^x dx$$

$$\int_0^1 \left(x/x - x(x^2) - [x^2 - (x^2)^2] \right) dx$$

$$\int_0^1 (x^4 - x^3) dx$$

$$\left. \frac{x^5}{5} - \frac{x^4}{4} \right|_0^1$$

$$= \frac{1}{5} - \frac{1}{4} = -\frac{1}{20}$$

L.H.S. = R.H.S.
proved.

$$\oint (Mdx + Ndy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy.$$

→ Green's Theorem.

~~Q.E.D.~~

Stokes theorem:-
Line integral of a vector $\vec{A}$ along a closed curve is equal to surface integral of curl of that vector.

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_S (\nabla \times \vec{A}) \cdot d\vec{s}$$

Line integral surface integral.

Proof:-

• surface integral formula:-
 $\left[\iint_S \vec{A} \cdot d\vec{s} \text{ or } \sum_{i=1}^n \vec{A} \cdot \Delta \vec{s} \right]$

• curl of vector $\vec{A} = \nabla \times \vec{A}$

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix}$$

or $\left[\nabla \times \vec{A} = \frac{d\vec{A}}{ds} \right]$

• Line integral = $\oint_C \vec{A} \cdot d\vec{r}$

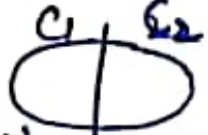
$$\vec{A} = A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}$$

$$d\vec{r} = (dx \hat{i} + dy \hat{j} + dz \hat{k})$$

$$\vec{A} \cdot d\vec{r} = (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \cdot (dx \hat{i} + dy \hat{j} + dz \hat{k})$$

$$\oint_C \vec{A} \cdot d\vec{r} = A_1 dx + A_2 dy + A_3 dz$$

$\oint_C \vec{A} \cdot d\vec{r}$ line integral along a closed curve.
Now,

$$\oint_C \vec{A} \cdot d\vec{r} = \oint_{C_1} \vec{A} \cdot d\vec{r} + \oint_{C_2} \vec{A} \cdot d\vec{r}$$




Similarly we can divide it upto n-curves.

$$\oint_C \vec{A} \cdot d\vec{r} = \oint_{C_1} \vec{A} \cdot d\vec{r} + \oint_{C_2} \vec{A} \cdot d\vec{r} + \dots + \oint_{C_n} \vec{A} \cdot d\vec{r}$$

$$\oint \vec{A} \cdot d\vec{r} = \sum_{i=1}^n \oint_{C_i} \vec{A} \cdot d\vec{r}$$

Multiply and divided by ΔS . $\nabla \times \vec{A} = \frac{\oint \vec{A} \cdot d\vec{r}}{\Delta S}$

$$= \sum_{i=1}^n \left(\frac{\oint_{C_i} \vec{A} \cdot d\vec{r}}{\Delta S} \right) \Delta S$$

$\Delta S \rightarrow$ curl of $\vec{A}$

$$\sum_{i=1}^n (\nabla \times \vec{A}) \Delta S \quad \text{which is curl of surface integral}$$

or

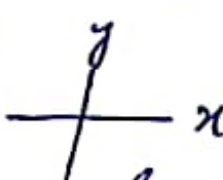
$$\oint \vec{A} \cdot d\vec{r} = \iint (\nabla \times \vec{A}) \cdot d\vec{S}$$

In limiting case $\sum_{i=1}^n \Rightarrow \iint$ and $\Delta S \rightarrow dS$.
proved.

Q. verify Stokes theorem for $\vec{A} = (2x-y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}$ where S is the upper half surface of sphere $x^2+y^2+z^2=1$ and C is its boundary.

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_S (\nabla \times \vec{A}) \cdot d\vec{S}$$

circle
 $\downarrow$ 2D
 eqn of circle:-
 $x^2 + y^2 = r^2$



sphere
 $\downarrow$ 3D

eqn of sphere:-
 $x^2 + y^2 + z^2 = r^2$
 $x^2 + y^2 + z^2 = 1$ $r=1$
 By taking upper half
 plane ans will be xy
 and $z=0$ Now
 $x^2 + y^2 + 0^2 = r^2$
 $x^2 + y^2 = r^2$ eqn of circle!



$$x = r \cos \theta, \quad y = r \sin \theta$$

$$x = \cos \theta, \quad y = \sin \theta$$

Now

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_S (\nabla \times \vec{A}) \cdot d\vec{S}$$

$$\oint_C ((2x-y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$\oint_C \vec{A} \cdot d\vec{r} = \int (2x-y) dx - yz^2 dy - y^2z dz \quad \text{--- (1)}$$

Now xy -plane put $z=0$ eq (1) becomes.

$$\oint_C \vec{A} \cdot d\vec{r} = \int (2x-y) dx \rightarrow \text{--- (2)}$$

$$x = r \cos \theta \Rightarrow \cos \theta$$

$$y = r \sin \theta \Rightarrow \sin \theta$$

$$x = \cos \theta$$

$$1 = -\sin \theta \frac{d\theta}{dx}$$

$$dx = -\sin \theta d\theta$$

Put value in (A) -

$$\oint (2 \cos \theta - \sin \theta) (-\sin \theta) d\theta$$

Circle's rotation in anticlockwise
0 to 2π so limit will be from 0 to 2π



$$= \int_0^{2\pi} (-2 \sin \theta \cos \theta + \sin^2 \theta) d\theta$$

$$= \int_0^{2\pi} (-\sin 2\theta + \frac{1 - \cos 2\theta}{2}) d\theta$$

$$= \int_0^{2\pi} -\sin 2\theta d\theta + \int_0^{2\pi} \frac{1}{2} d\theta - \int_0^{2\pi} \frac{\cos 2\theta}{2} d\theta$$

$$= \left[\frac{\cos 2\theta}{2} \right]_0^{2\pi} + \left[\frac{\theta}{2} \right]_0^{2\pi} - \left[\frac{\sin 2\theta}{2} \right]_0^{2\pi}$$

By putting limit

$$\oint \vec{A} \cdot d\vec{r} = \pi$$

Now
(integral)
surface
equal to
circle

R.H.S.

is in xy-plane which will
whole area and area of
Now $r=1$
 $\pi(1)^2 = \pi$

$$L.H.S = R.H.S.$$

$$\oint \vec{A} \cdot d\vec{V} = \iint_S (\nabla \times \vec{A}) \cdot d\vec{s} \rightarrow \text{verified}$$

Laplace transformations

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt.$$

Prove:

- $\mathcal{L}\{1\} = \frac{1}{s}$
- $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$
- $\mathcal{L}\{\cos at\} = \frac{s}{s^2+a^2}$
- $\mathcal{L}\{\sin at\} = \frac{a}{s^2+a^2}$
- $\mathcal{L}\{\cosh at\} = \frac{s}{s^2-a^2}$
- $\mathcal{L}\{\sinh at\} = \frac{a}{s^2-a^2}$
- $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$

① $\mathcal{L}\{1\} = \frac{1}{s}$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

$$\mathcal{L}\{1\} = \int_0^{\infty} e^{-st} (1) dt$$

$$\begin{aligned} & \int_0^{\infty} e^{-st} dt \Rightarrow \left. \frac{e^{-st}}{-s} \right|_0^{\infty} \\ & = \frac{e^{-s(\infty)}}{-s} - \left(\frac{e^{-s(0)}}{-s} \right) \\ & = \frac{e^{\infty}}{s} + \frac{e^0}{s} \Rightarrow 0 + \frac{1}{s} \end{aligned}$$

$$\mathcal{L}\{1\} = \frac{1}{s}$$

$$\textcircled{2} \mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

$$\therefore e^{\infty} = 0$$

$$e^0 = 1$$

$$\mathcal{L}\{e^{at}\} = \int_0^{\infty} e^{-st} e^{at} dt$$

$$\int_0^{\infty} e^{-(s-a)t} dt \Rightarrow \frac{e^{-(s-a)t}}{-(s-a)} \Big|_0^{\infty}$$

$$\frac{e^{-(s-a)\infty}}{-(s-a)} - \left(-\frac{e^{-(s-a)0}}{-(s-a)} \right)$$

$$\frac{e^{\infty}}{-(s-a)} + \frac{e^0}{(s-a)} \Rightarrow 0 + \frac{1}{(s-a)}$$

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

Trigonometric exponential formula

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}$$

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\textcircled{3} \mathcal{L}\{\cos at\} = \frac{s}{s^2 + a^2}$$

$$\mathcal{L}\left\{\frac{e^{iat} + e^{-iat}}{2}\right\} = \frac{1}{2} (\mathcal{L}\{e^{iat}\} + \mathcal{L}\{e^{-iat}\})$$

$$\frac{1}{2} \left[\frac{1}{s-ia} + \frac{1}{s-(-ia)} \right]$$

$$\frac{1}{2} \left[\frac{1}{s-ia} + \frac{1}{s+ia} \right] = \frac{1}{2} \left(\frac{(s+ia) + (s-ia)}{(s-ia)(s+ia)} \right)$$

$$\frac{1}{2} \frac{2s}{s^2 - i^2 a^2} = \frac{s}{s^2 + a^2}$$

$$\textcircled{4} \mathcal{L}\{\sin at\} = \frac{a}{s^2 + a^2}$$

$$= \mathcal{L} \left\{ \frac{e^{iat} - e^{-iat}}{2i} \right\} \quad \sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

$$= \frac{1}{2i} \left[\mathcal{L}\{e^{iat}\} - \mathcal{L}\{e^{-iat}\} \right]$$

$$= \frac{1}{2i} \left[\frac{1}{s-ia} - \frac{1}{s+ia} \right]$$

$$= \frac{1}{2i} \left[\frac{1}{s-ia} - \frac{1}{s+ia} \right]$$

$$= \frac{1}{2i} \left[\frac{(s+ia) - (s-ia)}{(s-ia)(s+ia)} \right]$$

$$\frac{1}{2i} \frac{s+ia - s+ia}{s^2 - i^2 a^2} = \frac{1}{2i} \frac{2ia}{s^2 + a^2}$$

$$= \frac{a^2}{s^2 + a^2} \quad \text{proved.}$$

$$\textcircled{5} \mathcal{L}\{ \cosh at \} = \frac{s}{s^2 - a^2}$$

$$\mathcal{L}\{ \cosh at \} = \mathcal{L}\left\{ \frac{e^{at} + e^{-at}}{2} \right\}$$

coshat

$$= \frac{1}{2} \left(\mathcal{L}\{ e^{at} \} + \mathcal{L}\{ e^{-at} \} \right)$$

$$= \frac{1}{2} \left(\frac{1}{s-a} + \frac{1}{s-(-a)} \right) = \frac{1}{2} \left(\frac{1}{s-a} + \frac{1}{s+a} \right)$$

$$= \frac{1}{2} \left(\frac{(s+a) + (s-a)}{(s-a)(s+a)} \right) = \frac{1}{2} \frac{2s}{s^2 - a^2}$$

$$= \frac{s}{s^2 - a^2}$$

$$\textcircled{6} \mathcal{L}\{ \sinh at \} = \frac{a}{s^2 - a^2}$$

$$\mathcal{L}\{ \sinh at \} = \mathcal{L}\left\{ \frac{e^{at} - e^{-at}}{2} \right\}$$

sinhat

$$= \frac{1}{2} \left(\mathcal{L}\{ e^{at} \} - \mathcal{L}\{ e^{-at} \} \right)$$

$$= \frac{1}{2} \left(\frac{1}{s-a} - \frac{1}{s-(-a)} \right)$$

$$= \frac{1}{2} \left(\frac{(s+a) - (s-a)}{(s-a)(s+a)} \right) = \frac{1}{2} \frac{2a}{s^2 - a^2}$$

$$= \frac{a}{s^2 - a^2}$$

$$\textcircled{7} \therefore \mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$

$$\mathcal{L}\{t^n\} = \int_0^{\infty} e^{-st} \overset{(E)}{t^n} \overset{(A)}{1} dt \rightarrow \text{ILATE}$$

$$\int_0^{\infty} t^n e^{-st} dt$$

Now integration by parts.

$$f(x) = t^n \rightarrow f'(x) = n t^{n-1}$$

$$g(x) = e^{-st} \rightarrow \int g(x) dx = \frac{e^{-st}}{-s}$$

Formula:

$$f(x)g(x) - \int \int g(x) f'(x) dx$$

$$t^n \frac{e^{-st}}{-s} \Big|_0^{\infty} - \int_0^{\infty} \frac{e^{-st}}{-s} n t^{n-1} dx$$

$$0 + \frac{n}{s} \int_0^{\infty} e^{-st} t^{n-1} dx$$

$$= \frac{n}{s} \int_0^{\infty} \underbrace{t^{n-1}} e^{-st} dx$$

again integration by parts.

$$f(x) = t^{n-1} \rightarrow f'(x) = (n-1) t^{n-2}$$

$$g(x) = e^{-st} \rightarrow \int g(x) dx = \frac{e^{-st}}{-s}$$

$$= \frac{n}{s} \left[f(x)g(x) - \int \int g(x) f'(x) dx \right]$$

$$\frac{n}{s} \left[t^{n-1} \frac{e^{-st}}{-s} \Big|_0^{\infty} - \int_0^{\infty} \frac{e^{-st}}{-s} (n-1) t^{n-2} dx \right]$$

$$\frac{n}{s} \left[0 + \frac{(n-1)}{s} \int_0^{\infty} e^{-st} t^{n-2} dx \right]$$

upto 'n' term

$$\frac{n}{s} \left(\frac{n-1}{s} \left(\frac{n-2}{s} \right) \dots \frac{1}{s} \mathcal{L}\{1\} \right)$$

$$\therefore \mathcal{L}\{1\} = \frac{1}{s}$$

$$\frac{n!}{s^n} \cdot \frac{1}{s} \Rightarrow \frac{n!}{s^{n+1}} \quad (n)(n-1)(n-2) \dots 1 = n!$$

Exercise - Find Laplace of following

Q# $t^2 - 16t + 17$

$$\mathcal{L}\{t^2 - 16t + 17\}$$

$$\mathcal{L}\{t^2\} - 16\mathcal{L}\{t\} + 17\mathcal{L}\{1\}$$

$$\therefore \mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$

$$\frac{2!}{s^{2+1}} - 16\left[\frac{1!}{s^{1+1}}\right] + 17\left[\frac{1}{s}\right] \quad \mathcal{L}\{1\} = \frac{1}{s}$$

$$\frac{2}{s^3} - \frac{16}{s^2} - \frac{17}{s}$$

Q#

$$e^{3t+5}$$

$$\mathcal{L}\{e^{3t} e^5\}$$

$$e^5 \mathcal{L}\{e^{3t}\}$$

$$e^5 \frac{1}{s-3} \Rightarrow \frac{e^5}{s-3}$$

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

Q# $\sin(7t + 4)$

$$\mathcal{L}\{\sin(7t + 4)\}$$

$$\begin{aligned} \sin(\alpha + \beta) &= \sin\alpha \cos\beta + \cos\alpha \sin\beta \\ \sin(\alpha - \beta) &= \sin\alpha \cos\beta - \cos\alpha \sin\beta \\ \cos(\alpha + \beta) &= \cos\alpha \cos\beta - \sin\alpha \sin\beta \\ \cos(\alpha - \beta) &= \cos\alpha \cos\beta + \sin\alpha \sin\beta \end{aligned}$$

$$\mathcal{L}(\sin 7t + \cos 4 + \cos 7t + \sin 4)$$

$$\therefore \mathcal{L}\{\cos at\} = \frac{s}{s^2 + a^2}$$

$$\mathcal{L}\{\sin 7t\} \cos 4 + \mathcal{L}\{\cos 7t\} \sin 4$$

$$\frac{7}{s^2 + (7)^2} \cos 4 + \frac{s}{s^2 + (7)^2} \sin 4$$

$$\frac{7 \cos 4}{s^2 + 49} + \frac{s \sin 4}{s^2 + 49}$$

$$Q\# \cos(at+b)$$

$$\mathcal{L}\{\cos at \cos b - \sin at \sin b\}$$

$$\mathcal{L}\{\cos at\} \cos b - \mathcal{L}\{\sin at\} \sin b$$

$$\frac{s}{s^2 + a^2} \cos b - \frac{a}{s^2 + a^2} \sin b$$

$$Q\# (t^3 - 1)e^{-2t}$$

$$\mathcal{L}\{(t^3 - 1)e^{-2t}\}$$

$$\mathcal{L}\{t^3 e^{-2t} - e^{-2t}\}$$

$$\mathcal{L}\{t^3 e^{-2t}\} - \mathcal{L}\{e^{-2t}\}$$

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$

$$\frac{3!}{(s - (-2))^{3+1}} - \frac{1}{s - (-2)}$$

$$\frac{6}{(s+2)^4} - \frac{1}{s+2}$$

Q# $e^{3t}(t^3 + \sin 2t)$

$$\mathcal{L}(e^{3t}t^3 + e^{3t}\sin 2t)$$

$$\mathcal{L}\{e^{3t}t^3\} + \mathcal{L}\{e^{3t}\sin 2t\}$$

$$\frac{3!}{(s-3)^{3+1}} + \left(\frac{2}{s^2 + (2)^2}\right)e^{3t}$$

$$\frac{6}{(s-3)^4} + \frac{2}{(s-3)^2 + 4}$$

Q# $e^{3t} \cosh 4t$

$$\mathcal{L}\{e^{3t} \cosh 4t\}$$

$$\cosh at = \frac{s}{s^2 - a^2}$$

$$\frac{s-3}{(s-3)^2 - (4)^2}$$

$$\Rightarrow \frac{s-3}{(s-3)^2 - 16}$$

Q# $e^{-t} \sin 2t$

$$\mathcal{L}\{e^{-t} \sin 2t\}$$

$$e^{-t} \frac{2}{s^2 + 4}$$

$$\frac{2}{(s-0)^2 + 4}$$

$$\mathcal{L}\{\sin at\} = \frac{a}{s^2 + a^2}$$

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

Fourier transformation:-

$$F[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-i\omega x} f(x) dx.$$

Q# Find Fourier transformation of
OR Find Fourier transform of

$$f(x) = \begin{cases} 1 & \text{if } x \geq 1 \\ 0 & \text{if } x \leq 1 \end{cases}$$

$$F[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i\omega x} f(x) dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_1^{\infty} e^{-i\omega x} (1) dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_1^{\infty} e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{-i\omega x}}{-i\omega} \right]_1^{\infty}$$

$$= \frac{1}{\sqrt{2\pi}} \left(\frac{e^{-i\omega(\infty)} - e^{-i\omega(-1)}}{-i\omega} \right)$$

$$\frac{1}{\sqrt{2\pi}} \frac{e^{-i\omega} - e^{+i\omega}}{-i\omega} \Rightarrow \frac{1}{\sqrt{2\pi}} \left(\frac{e^{+i\omega} - e^{-i\omega}}{i\omega} \right)$$

$$\frac{1}{\sqrt{2\pi}} \left(\frac{e^{i\omega} - e^{-i\omega}}{i} \right) \frac{1}{\omega}$$

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

$$\sin \omega = \frac{e^{i\omega} - e^{-i\omega}}{2i}$$

$$\frac{1}{\sqrt{2\pi}} \cdot \frac{2}{2} \left(\frac{e^{i\omega} - e^{-i\omega}}{i} \right) \frac{1}{\omega}$$

$$\frac{1}{\sqrt{2\pi}} \cdot 2 \frac{\sin \omega}{\omega} \Rightarrow \frac{2}{\sqrt{2\pi}} \frac{\sin \omega}{\omega}$$

Q. Find limit

Fourier sine transformation:-

$$F_s[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin wx dx$$

Fourier cosine transformation:-

$$F_c[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos wx dx.$$

Q# Find Fourier sine transformation of e^{-x}

$$F_s[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin wx dx$$

$$\sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-x} \sin wx dx$$

Ⓔ Ⓙ Integration by parts.

ILATE

$$\sqrt{\frac{2}{\pi}} \int_0^{\infty} \underbrace{\sin wx}_{\text{I}} \underbrace{e^{-x}}_{\text{L}} dx \quad \text{--- (A)}$$

$$f(x) = \sin wx$$

$$g(x) = e^{-x}$$

→

$$f'(x) = \cos wx(w)$$

$$\int g(x) dx = \frac{e^{-x}}{-1}$$

$$I = \int_0^{\infty} \sin \omega x e^{-x} dx$$

$$f(x) \int g(x) - \int \int g(x) f'(x) dx \rightarrow \text{formula}$$

$$I = \sin \omega x \frac{e^{-x}}{-1} \Big|_0^{\infty} - \int_0^{\infty} \frac{e^{-x}}{-1} \omega \cos \omega x dx$$

$$I = 0 + \omega \int_0^{\infty} e^{-x} \cos \omega x dx$$

$$I = \omega \int_0^{\infty} \cos \omega x e^{-x} dx$$

again integration
by parts

$$f(x) = \cos \omega x \rightarrow f'(x) = -\sin \omega x (\omega)$$

$$g(x) = e^{-x} \rightarrow \int g(x) dx = \frac{e^{-x}}{-1}$$

$$I = \omega \left[f(x) \int g(x) - \int \int g(x) f'(x) dx \right]$$

$$\omega \left[\cos \omega x \frac{e^{-x}}{-1} - \int_0^{\infty} \frac{e^{-x}}{-1} \omega \sin \omega x dx \right]$$

$$I = \omega \left[\infty - \cos \omega(0) \frac{e^{-0}}{-1} - \omega \int_0^{\infty} \frac{e^{-x} \sin \omega x dx}{I} \right]$$

$$I = \omega [1 - \omega I] \quad \because \cos 0 = 1, e^0 = 1$$

$$I = \omega - \omega^2 I$$

$$I + \omega^2 I = \omega$$

$$I(1 + \omega^2) = \omega$$

$$\Rightarrow I = \frac{\omega}{1 + \omega^2}$$

Put value of R in (A)

$$F_s [e^{-\pi}] = \sqrt{\frac{2}{\pi}} \frac{W}{1+W^2}$$

Q# Find " " " " " " " " " " " "

If $f(x)$ and $f'(x)$ both vanish as $x \rightarrow \infty$, then show that

$$F_c[f''(x)] = -\omega^2 F_c[f(x)] - \sqrt{\frac{2}{\pi}} f'(0)$$

$$F_c[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos \omega x \, d\omega$$

$$F_c[f''(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f''(x) \cos \omega x \, d\omega$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} \cos \omega x \, f''(x) \, dx$$

$$f(x) = \cos \omega x \rightarrow f'(x) = -\omega \sin \omega x$$

$$g(x) = f'(x) \rightarrow \int g(x) \, dx = f'(x)$$

$$= \sqrt{\frac{2}{\pi}} \left[\int_0^{\infty} f(x) g(x) \, dx - \int_0^{\infty} g(x) f'(x) \, dx \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[\cos \omega x \, f'(x) \right]_0^{\infty} - \int_0^{\infty} f'(x) - \omega \sin \omega x \, dx$$

$$= \sqrt{\frac{2}{\pi}} \left[\infty - \cos \omega(0) f'(0) + \omega \int_0^{\infty} \sin \omega x \, f'(x) \, dx \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[-f'(0) + \omega \left[f(x) g(x) - \int_0^{\infty} g(x) f'(x) \, dx \right] \right]$$

$$f(x) = \sin \omega x \Rightarrow f'(x) = \omega \cos \omega x$$

$$g(x) = f'(x) \Rightarrow \int g(x) = f(x)$$

$$= \sqrt{\frac{2}{\pi}} \left[-f'(0) + \omega \int_0^{\infty} \sin \omega x f(x) dx - \int_0^{\infty} f(x) \omega \cos \omega x dx \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[-f'(0) + \omega \left[0 - \omega \int_0^{\infty} f(x) \cos \omega x dx \right] \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[-f'(0) - \omega^2 \int_0^{\infty} f(x) \cos \omega x dx \right]$$

$$= \sqrt{\frac{2}{\pi}} f'(0) - \omega^2 \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos \omega x dx$$

$$\mathcal{F}_c \{ f(x) \}$$

$$\mathcal{F}_c \{ f''(x) \} = -\sqrt{\frac{2}{\pi}} f'(0) - \omega^2 \mathcal{F}_c \{ f(x) \}$$

7.1 Z-Transform

Definition 7.1 — Z-Transform. Let $\{u_n\}$ be a sequence defined for discrete values $n = 0, 1, 2, \dots$ then, the Z-transform of sequence $\{u_n\}$ is denoted by $Z(u_n)$ and defined as

$$Z(u_n) = \sum_{n=-\infty}^{\infty} u_n z^{-n} = U(z)$$

Provided the infinite series is convergent and U is a function of complex number z .

■ **Example 7.1** Find the Z-transform of the following sequence:

(i) $\{15, 10, 7, 4, 1, -1, 0, 3, 6\}$ (ii) $\left\{\frac{1}{2^n}\right\}, -2 \leq n < 4$ (iii) $\left\{\frac{1}{3^n}\right\}$

Solution:

$$(i) \quad Z\{u_n\} = \sum_{n=-\infty}^{\infty} u_n z^{-n}$$

$$Z\{u_n\} = 15z^4 + 10z^3 + 7z^2 + 4z + 1 - \frac{1}{z} + \frac{0}{z^2} + \frac{3}{z^3} + \frac{6}{z^4}.$$

$$(ii) \quad Z\{u_n\} = \sum_{n=-2}^4 \frac{1}{2^n} z^{-n} = 4z^2 + 2z + 1 + \frac{1}{2z} + \frac{1}{4z^2} + \frac{1}{8z^3} + \frac{1}{16z^4}.$$

$$(iii) \quad Z\{u_n\} = \sum_{n=-\infty}^{\infty} \frac{1}{3^n} z^{-n} = \dots + 9z^2 + 3z + 1 + \frac{1}{3z} + \frac{1}{9z^2} + \frac{1}{27z^3} + \dots.$$

■ **Example 7.2** Find the Z-transform of $\left(\frac{a^n}{n!}\right)$, $n \geq 0$.

Solution:

$$\begin{aligned} Z\left\{\frac{a^n}{n!}\right\} &= \sum_{n=0}^{\infty} \frac{a^n}{n!} z^{-n} = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{a}{z}\right)^n \\ &= 1 + \frac{a}{z} + \frac{1}{2!} \left(\frac{a}{z}\right)^2 + \frac{1}{3!} \left(\frac{a}{z}\right)^3 + \dots = e^{\frac{a}{z}}. \end{aligned}$$

■ **Example 7.3** Find the Z-transform of $\left(\frac{a^n}{n}\right)$, $n \geq 1$.

Solution:

$$\begin{aligned} Z\left\{\frac{a^n}{n}\right\} &= \sum_{n=1}^{\infty} \frac{a^n}{n} z^{-n} = \sum_{n=1}^{\infty} \frac{1}{n} \left(\frac{a}{z}\right)^n \\ &= \frac{a}{z} + \frac{1}{2} \left(\frac{a}{z}\right)^2 + \frac{1}{3} \left(\frac{a}{z}\right)^3 + \dots \\ &= -\log\left(1 - \frac{a}{z}\right). \end{aligned}$$